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Kiran Mandal

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Fundamental Research(TIFR),
Hyderabad, INDIA

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Date of Birth: 06.04.2003 (Age: 23)
Male , Unmarried

Research experience

- Studied effects of quantum fluctuations for **Cavity-Magnon** system for Heisenberg **ferro-and antiferromagnets** coupled to cavity photons, where we investigated the behaviour of order parameter, phase transition, dynamical structure factor and how strong light-matter hybridization renormalizes the quasiparticle dispersion of this system.
- Studied the **Cavity-Triplon** coupling in **Topological** quantum paramagnetic spin ladder and it's effect on energy modes, specially on localized edge modes and local density of states etc.
- Worked on **Quantum Refrigerator** based on **parametric cooling** of Simple Harmonic Oscillator.
- Have done an extensive literature review on **Quantum Circuits** including **LC resonator**, Josephson Parametric amplifier (**JPA**) , Rabi Oscillation , **Free Inductive Decay and spin echo**, **Jaynes-Cummings** Model, **Lindblad Master Equation** and **Purcell effect**.

Experimental Research Experience

- Developed monolayer molybdenum disulphide (**MoS₂**) via two zone chemical vapour deposition (CVD).
- Explored formation of Magneto-vortex in Permalloy Nano-disks .

Education:

- **Tata Institute of Fundamental Research** Hyderabad, India
*Master's in science - Physics; current **CGPA: 8.02*** August 2024 - At present
Courses : *Advanced Quantum Mechanics , Quantum Optics , Condensed Matter physics, Reading Project on Quantum Circuits, Quantum Transport, Elements of Magnetism, Numerical Methods, Statistical Mechanics, Laser Physics etc.*
- **Bangabasi College, University of Calcutta** Kolkata , India
*Bachelor's in Science - Physics ; **CGPA : 7.61*** August 2020 - July 2023
Courses: *Quantum Mechanics , Classical Mechanics, Nano Physics, Advance Mathematical Methods, Electronics , Electromagnetic - Theory etc.*
- **West Bengal Council of Higher Secondary Education** West Bengal, India
Science stream ; Marks ; 84% May 2018-August 2020

Publication/Preprints :

No publications yet

Poster Presentation and seminars :

- Selected to present a poster at the upcoming conference “Quantum Peaks 2026 A Himalayan Summit on Coherence in Light and Matter”.
(IIT-Mandi, India, 14-18 Nov, 2026)

Achievements:

- All India Rank 30 in JEST(joint entrance screening test) out of ~8000 candidates.
- All India Rank 569 in IIT-JAM out of ~ 13000 candidates
- Selected in IIT's, IISER's, NISER, IUCAA through interview and screening test.(declined)

Volunteering and Extracurriculars:

- Optica : Participated in student outreach program.
- Organised and hosted weekly seminars at undergraduate college, featuring renowned Indian physicist like Nabakumar Mandal(TIFR) , Krishnendu Sengupta(IACS) and Rajarshi Raut. (Kolkata, West Bengal)

Skills:

- Python(Numpy , sympy)
- Qiskit (Quantum Diagonalization Algorithms)
Variational Quantum Eigensolver,
Quantum Krylov Method
Sample Based Quantum Diagonalization
- Mathematica
- Matlab
- Latex
- Microsoft office

Lab experiment skills:

- CVD(chemical vapour deposition)
- Raman Spectroscopy
- Photolithography

Languages: Bengali(Native), English(Professional),
Hindi

Recommendations:

1. Dr. Darshan G. Joshi

Position: Assistant Professor, Tata Institute of Fundamental Research, Hyderabad (TIFRH).

Email: djoshi@tifrh.res.in

2. Dr. Partha Ghosh

Position: Associate Professor, Bangabasi College, University of Calcutta.

Email: ghosh_partha25@rediffmail.com

3. Dr. Sreenath K. Manikandan

Position: Assistant Professor, Tata Institute of Fundamental research, Hyderabad (TIFRH).

Email: skm@tifrh.res.in